

BZTel - Monty Mobile SMPP Integration Guide

Technical Reference & Troubleshooting Guide for IT Support Call
Date: July 23, 2026

1. Executive Connection Summary

A diagnostic audit was conducted from the active BZTel AWS EC2 server to troubleshoot the SMPP connection issues with the Monty Mobile SMSC gateway. The investigation reveals two distinct issues:

- **AWS EC2 Server Public IP:** 52.51.186.75 (Confirmed static outbound IP)
- **Documented IP Discrepancy:** The initial meeting brief listed BZTel AWS IP as 176.34.210.141. It is highly likely Monty Mobile IT whitelisted the old IP instead of the active IP 52.51.186.75.

Gateway Status Breakdown:

- **Nigeria Gateway (154.113.5.24:9013):** BLOCKED (Network-Level Timeout)
All TCP handshakes from the AWS server to port 9013 fail immediately. The network-level firewall on Monty Mobile's side has not yet whitelisted our IP 52.51.186.75.
- **Global/Test Gateway (185.135.128.117:7500):** REJECTED (Application-Level Drop)
TCP connection completes, but the Monty Mobile SMSC immediately closes the socket when we send an SMPP bind_transceiver or bind_transmitter command. This occurs because the SMSC daemon rejects the credentials from our source IP 52.51.186.75.

2. Step-by-Step Connection & Troubleshooting Tests

To verify status updates with the Monty Mobile IT team during the call, perform the following steps in sequence. All tests must be initiated from the BZTel AWS EC2 server.

Step 1: SSH into the AWS EC2 Instance

From your local machine (inside the project root folder containing the bztel-key.pem file), execute:

```
ssh -i .\bztel-key.pem -o StrictHostKeyChecking=no ubuntu@52.51.186.75
```

Step 2: Run Network-Level Reachability Tests (Netcat)

Verify if the network-level firewall has been updated by Monty Mobile to allow TCP traffic.

- **Test connection to the Nigeria Gateway:**

```
nc -zv -w 5 154.113.5.24 9013
```

- Expected failure: nc: connect to 154.113.5.24 port 9013 (tcp) timed out
- Expected success: Connection to 154.113.5.24 9013 port [tcp/*] succeeded!

Step 3: Run SMPP Protocol Bind Tests

Once Step 2 succeeds, run the automated test scripts on the server to verify the SMPP handshake.

- **Basic Dual-Account Connection Test:**

Attempts to bind BZTltms (Transactional) and BZTIPrmo (Promotional) using transceiver mode.

```
node test_smpp.js
```

- **Password Variant Testing:**

Tests 5 password variants (including URL-decoded, truncated, and raw formats) to rule out format mismatches.

```
node test_pwds.js
```

- **Comprehensive Parameter Matrix Testing:**

Tests 10 combinations of bind modes (transceiver, transmitter, receiver), SMPP versions (3.3, 3.4), and system types.

```
node test_comprehensive.js
```

Step 4: Update Gateway Environment Configuration

Once Monty Mobile whitelists the IP and the tests succeed, update the server configuration file.

- **Open .env file:** nano .env

- **Update Host and Port:** Set SMPP_HOST=154.113.5.24 and SMPP_PORT=9013

- **Save changes:** Press Ctrl + O, then Enter, then Ctrl + X

Step 5: Restart and Monitor the Production Worker

Restart the background worker process using PM2 to apply the configuration and start dispatching the queue.

```
pm2 restart bztel-worker --update-env
```

Monitor logs in real-time:

```
pm2 logs bztel-worker
```

3. Troubleshooting Request & Action Items

Present the following instructions to Monty Mobile IT Support during the call:

- **Update Whitelist IP:** Replace old whitelisted IP 176.34.210.141 with current active server IP 52.51.186.75.
- **Nigeria Gateway Firewall:** Permit inbound TCP traffic from 52.51.186.75 to port 9013 on gateway 154.113.5.24.
- **SMSC Client Mapping:** Verify that BZTltms and BZTIPrmo accounts are associated with IP 52.51.186.75 in the SMSC configuration.